Real-Time Traffic Accident

Severity Prediction

A Real-Time Big Data Pipeline using Apache Kafka & Apache Spark

ITCS 6190 — Cloud Computing for Data Analysis | Spring 2026 | UNC Charlotte

7.73M	46	49	7 Yrs
Records	Columns	US States	2016–2023

Meet The Team

Real-Time Traffic Accident Severity Prediction

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Team of 4 | All members contribute to the full pipeline

Problem Statement

Current traffic systems are entirely REACTIVE — by the time a warning reaches drivers, they are already stuck in congestion.

01

Late Detection

Accidents reported minutes after they occur. Following vehicles approach unaware, worsening congestion and increasing secondary accident risk.

02

No Automated Severity Assessment

No system instantly classifies whether an accident needs a minor response or a full emergency road closure. Done manually — slow and inconsistent.

03

No Pattern-Based Hotspot Intel

Traffic centers cannot instantly query which roads are most dangerous under current weather and time conditions. Requires hours of manual analysis.

04

Data Processing Bottleneck

Millions of records accumulating continuously. Traditional batch processing systems are far too slow for real-time decision making needs.

Dataset Overview

US Accidents Dataset (2016–2023)

File	US_Accidents_March23.csv
Source	Kaggle — Sobhan Moosavi
Total Rows	7,728,394 records
Total Cols	46 columns
Coverage	49 US States
Time Period	Feb 2016 – March 2023
File Size	~3.2 GB
Collection	Multiple real-time APIs

46 Columns — 6 Attribute Groups

3 — **Identity & Core**
ID, Source, Severity

2 — **Temporal**
Start_Time, End_Time

9 — **Location**
Lat, Lng, Street, City, County, State, Zipcode

10 — **Weather**
Temperature, Visibility, Wind Speed, Humidity, Precipitation...

13 — **Road Features**
Junction, Traffic_Signal, Crossing, Bump, Railway...

4 — **Day / Night**
Sunrise_Sunset, Civil_Twilight, Nautical_Twilight...

Severity Levels — The Core Target Variable

Severity = How badly the accident disrupted traffic flow (NOT about injury level)

1
Very Minor

Two cars lightly tap and pull to the side immediately. Traffic barely notices. Road clears in minutes.

Duration: 2–5 min

Response: None needed

Rarest

2
Light

Car breaks down in right lane. Small slowdown, police arrive, tow truck dispatched. Clears soon.

Duration: 10–30 min

Response: Police + Tow Truck

MOST COMMON

3
Serious

Truck hits car. Full lane blocked for extended period. Traffic backs up for miles. Detours begin.

Duration: 45min–2hr

Response: Police + Ambulance + Tow

Common

4
Severe

Major pile-up. Highway completely closed for hours. Emergency helicopters. News coverage live.

Duration: 3–6+ hrs

Response: Full Emergency Response

Only ~2.65%

Three Layers of Analytics

01

Descriptive Analytics

What is happening RIGHT NOW?

Tool: Spark SQL

- Top 10 cities by avg severity last 15 mins
- Weather conditions causing Severity 3 and 4
- Rush hour vs off-peak severity comparison
- State-level live incident rate summaries

02

Diagnostic Analytics

WHY is it happening?

Tool: Spark Structured Streaming

- Detect dangerous combinations of conditions
- Low visibility + junction + rush hour = high risk
- Rolling 5 / 15 / 60 min window aggregations
- Flag compound patterns before full escalation

03

Predictive Analytics

What will happen NEXT?

Tool: MLlib — Random Forest

- Predict Severity 1 to 4 in milliseconds
- Trained on 7.73M historical accidents
- 12 weather road and time features as input
- Auto-generate alerts for Severity 3 and 4

Machine Learning — Random Forest Classifier

Why Machine Learning?

Severity depends on 12 factors interacting simultaneously. No single rule captures all combinations. ML learns patterns from 7.73M real accidents and applies that knowledge instantly to every new incoming stream record.

Why Random Forest?

- ✓ Handles 4-class Severity prediction natively
- ✓ Works with numbers, categories and True/False flags
- ✓ Robust to missing values in the dataset
- ✓ Provides ranked feature importance after training

12 Input Features — What the Model Looks At

Temperature(F)
Cold = ice risk on road

Visibility(mi)
Low = drivers cannot react

Wind Speed(mph)
High = lose vehicle control

Humidity(%)
High = wet slippery roads

Precipitation(in)
Rain or snow amount

Weather Condition
Snow vs rain vs fog

Junction
Intersections = highest risk

Traffic Signal
Controlled vs uncontrolled

Sunrise/Sunset
Night = more severe

Hour of Day
Rush hour 7–9AM, 4–6PM

Day of Week
Weekday vs weekend

Distance(mi)
How much road is blocked

How the Model Works in Real Time

Time-Based Split Strategy — NOT Random Split (respects temporal ordering of accident data)

TRAINING SET — 2016 to 2021 (80% = 6.18 Million records)	TEST 2022–23 20%

Model learns from older data. Test set simulates real time — model has NEVER seen these records before. Prevents data leakage.

PHASE 1 — Offline Training

1. Load 2016–2021 accident records
2. Engineer 12 features from raw columns
3. Train Random Forest Classifier on training set
4. Evaluate Accuracy and F1 on 2022–2023 test set
5. Save trained model to disk — ready to deploy

PHASE 2 — Online Real-Time Inference

1. New accident record streams in via Kafka live
2. Spark extracts its 12 features instantly
3. Saved model predicts Severity in milliseconds
4. Severity 3 or 4 triggers automatic alert output
5. No retraining — model applies pre-learned knowledge

System Architecture — Full Pipeline

DATA SOURCE

US_Accidents_March23.csv

7.73M rows · 46 columns

AWS S3 / Kaggle

→

KAFKA PRODUCER

TrafficDataGenerator.py

Row by row CSV reader

JSON to Kafka topic

→

SPARK STREAMING

StructuredStreaming.py

5/15/60 min windows

10-min watermark

→

OUTPUT SINK

Alerts Parquet files

Aggregates CSV

Partitioned by State

SPARK SQL

SparkSQL.py

Top 10 hotspot queries

Weather-severity analysis

MLlib MODEL

RandomForestClassifier

12 features input

Severity 1–4 output

EVALUATION

Accuracy + F1-Score

Confusion Matrix

Feature Importance

Spark processes Spark SQL queries and ML predictions simultaneously on the same stream

Evaluation Metrics — How We Know It Works

Accuracy

Definition:

Out of all Severity predictions made, how many did the model get correct overall across all 4 classes.

Why we use it:

Quick overall performance snapshot. Easy to understand and communicate to non-technical stakeholders.

Note: Can be misleading — model always predicting Severity 2 still looks accurate due to class imbalance in dataset.

Weighted F1-Score

Definition:

Balances Precision and Recall for each Severity class individually, then combines weighted by class frequency.

Why we use it:

Primary metric. Handles class imbalance — penalises ignoring rare Severity 4 events. Far more honest than accuracy.

Note: Severity 4 is only 2.65% of data. Plain accuracy would hide the model completely failing on this critical class.

Confusion Matrix

Definition:

Table showing how many predictions landed in each Severity class — correct and incorrect — across all combinations.

Why we use it:

Reveals exactly where the model makes mistakes. Shows if Severity 4 is ever misclassified as a lower severity.

Note: Missing a Severity 4 prediction means suppressing an emergency alert. Most critical failure mode to check.

Future Upgrades — Next Steps

01 — Live Weather API

Connect real-time weather to predict dangerous conditions BEFORE any accident is reported. True prevention over reaction.

02 — Visual Map Dashboard

Live color-coded US map — green, yellow, red severity zones. Updates every 15 minutes. Usable by non-technical traffic managers.

03 — SMS and Email Alerts

Auto-send notifications when Severity 3 or 4 detected. Authorities warned instantly without manually checking output files.

04 — State-Level Partitioning

Separate Kafka topic per state. All 49 states processed in parallel. Eliminates bottlenecks during high-volume periods.

05 — Auto Model Retraining

Monthly scheduled retraining on new accident data. Old vs new model compared. Better model auto-deployed without manual intervention.

06 — Context Data Integration

Add public events, road construction, school schedules. System understands game-day vs normal — same road, completely different risk.

Conclusion

What We Built

A real-time big data pipeline streaming 7.73 million US accident records through Apache Kafka and Spark — detecting high-severity incidents, running live SQL hotspot analysis, and predicting Severity 1 through 4 using Random Forest trained on 12 weather, road, and time features — all within 15-minute windows.

Problem	Solution	Dataset	Model
Reactive systems warn drivers too late	Real-time streaming + ML severity prediction	7.73M records · 46 cols · 49 States	Random Forest — 12 features → Severity 1–4

This project shifts traffic accident management from reactive response to proactive intelligence.